Which Way Does Fairness Go? The Selection Rate Predicts Whether an Attribute-Blind Constraint Gives or Takes

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***Abstract*—A group-fairness constraint requires two groups' selection rates to be close, but it does not say how the gap should close: the disadvantaged group can be lifted, or the advantaged group can be pulled down. Both outcomes satisfy the constraint, and the metric that certifies it reports the same number in each case, which means a deployment team cannot tell from its own dashboard which of the two it is about to cause. We show that the direction is predictable in advance from a quantity every deploying organisation already has: the rate at which its current model says yes, read against an empirically observed crossover near 0.54. Across 39 independent populations covering six decision domains and seven data sources, an attribute-blind parity constraint withdraws favourable decisions when the baseline selection rate sits below the crossover, and extends them when it sits above; on UCI Adult six in-processing methods each shrink the pool by 7.9–22.1%, while on mortgage lending the identical constraint grows it. The claim is *sealed* rather than fitted: with the rule, the populations and the pass mark committed to version control before any run, it predicted the direction on 9 of 10 populations never previously measured, while the best constant reached 6. An earlier sealed attempt that carried an in-sample refinement failed at four of eight, and we report both outcomes because the difference between them is what a sealed test exists to expose. The relationship survives a change in how the rate is varied, a twenty-five-fold range of constraint tolerance, a boosted-tree learner and a protected attribute swapped from sex to race, but it disappears under post-processing ($r = -0.024$ against $+0.585$), which reads the protected attribute at decision time. That boundary is where contemporaneous theory says the direction stops being distributional and becomes determined, and the theory's prediction for that regime holds in 27 of 27 populations, nine pre-registered. The practical outcome is a procedure: when the deployed model's viable operating band is wide enough — which the procedure itself checks first — a team can locate its own crossover by sweeping that model's threshold, using no new data.**

***Index Terms*—algorithmic fairness, levelling down, demographic parity, in-processing, selection rate, disparate impact**

I. Introduction

A fairness constraint says that two groups' rates should be close, but it does not say how that closeness should be reached. The gap can close because the disadvantaged group is lifted, or because the advantaged group is pulled down, and while both directions satisfy the constraint, only one of them is what anyone imposing it actually wants.

This ambiguity is already well established. Mittelstadt *et al.* [2] made levelling down the centre of a well-known critique; Maheshwari *et al.* [4] observe that it "often goes unnoticed in the overall performance of the model"; and Ferry *et al.* [3] built an auditing framework precisely because an aggregate score reports neither how many people were affected nor in which direction. We take all of that as given.

What the literature leaves open is the question a deploying team actually faces: **when** does each direction happen? A team cannot act on "this sometimes occurs", but it can act on a rule that says which way its own system will move before it moves, which is what this paper provides and tests.

A. Scope, stated first

This is a paper about **attribute-blind in-processing** constraints, meaning the regime in which the protected attribute is unavailable at decision time. Many deployed systems sit there, because per-group thresholds are widely read as disparate treatment on their face [10], [13], which rules out the attribute-aware alternatives in most regulated settings. Under post-processing, which does read the attribute, the effect we report disappears (Section VII). We treat this not as a weakness of the finding but as the boundary the theory draws, and we verify it from both sides.

B. Contributions

1) **An empirical characterisation of a conditionality that theory predicts.** Contemporaneous theory [5] establishes that in the attribute-blind regime the direction is distribution-dependent. It contains no experiments and its conditions are ordering relations on score regions rather than measurable quantities. We supply the measurement across 37 populations and six domains, and find its stated conditions — which are *sufficient, not necessary* — cannot be evaluated on any of 26 real arms, for a structural reason given in Section VII. A relaxed form of the same ordering agrees with the measured

direction on 24 of 26. The direction rule itself is *sealed*: committed before ten populations this work had never measured, it scored 9 of 10 against a best constant of 6 (Section VIII-D).
2) **Separation of the predictor from its confound.** Varying the selection rate by moving a label threshold cannot distinguish "the rate predicts the direction" from "task difficulty does". Holding the task exactly fixed and moving only the decision rule separates them. We claim a *predictive* relationship within a population under the tested regime, not a causal mechanism: no pooled slope transfers across populations (Section VIII).
3) **A procedure, not an observation.** The same design lets a practitioner locate their own crossover on a deployed model. Four independent estimates of the mortgage crossover agree.
4) **Boundaries stated explicitly rather than left for review to find**, including a computable test for when the procedure cannot be used at all.

II. Related Work

Whether levelling down is universal. A 2026 theory paper [5] proves in a population-level Bayes framework that the answer depends on the deployment regime. Where the protected attribute is available at decision time, fairness "necessarily (weakly) improves outcomes for the disadvantaged group and (weakly) worsens outcomes for the advantaged group". Where it is not — the attribute-blind regime, which is ours — "the impact of fairness is distribution-dependent: fairness can benefit or harm either group . . . leading to either leveling up or leveling down." **We therefore do not claim the conditionality itself.**

Levelling down, and the remedy. Mittelstadt *et al.* [2] argue that fairness interventions frequently equalise by degrading the better-off group, and *also propose the remedy*: minimum rate constraints requiring "that every group has, at least, a minimal selection rate, precision, or recall". We claim neither the observation nor the remedy as novel. Our contribution relative to that work is narrow and checkable: a scope condition they do not have; in-processing rather than post-processing, so no model reads the attribute when predicting; a population-level floor stacked *with* parity rather than a per-group floor replacing it; and held-out replication where they report training-set results.

Auditing individual impact. Ferry *et al.* [3] propose a framework whose first three dimensions are impact size, change direction and decision rates. That is the decomposition we use throughout, and we claim no part of it. What we add is what those axes report at scale: the change-direction dimension is not a property of the method being audited but of the population it is applied to.

The shape of the optimal fair classifier. Corbett-Davies *et al.* [10] show that for several fairness definitions the optimal constrained classifier takes the form of group-specific risk thresholds. This gives a *bound* rather than another baseline: it predicts that levelling down is not an artifact of the reduction's search, because the optimal constrained classifier withdraws in the same way.

Remedies that avoid harm. Minimax group fairness [11] minimises the worst group's error rather than equalising anything, and is the natural alternative to a parity constraint for a team whose concern is harm rather than equality. We benchmark against it rather than only against the unconstrained reduction.

The selection-rate axis. The nearest prior work is Goethals *et al.* [8], which studies the cost of fairness as a function of available budget and reports that "the level of available resources significantly influences this cost, a factor overlooked in previous evaluations". The difference is structural: in their setting positive decisions are a fixed resource, so the total is constant by construction and the directional effect we measure cannot arise.

Fairness gerrymandering and dataset monoculture. Kearns *et al.* [6] show constraints on marginal groups can be satisfied while subgroups are badly treated. Ding *et al.* [7] argue the field should stop drawing conclusions from UCI Adult and supply ACS-derived replacements. Notably they also *build the knob this paper turns*: the $50,000 threshold "was chosen so that this dataset can serve as a replacement to UCI Adult, but we also offer datasets with other income cutoffs". They do not connect it to the direction of a fairness intervention — their text contains no occurrence of "selection rate", "acceptance rate" or "leveling down" — so the instrument existed and the question was not asked of it. We replicate across their populations and then find that is *not sufficient*, because they share one survey instrument.

III. Setup and Method

Data. Thirty-nine independent populations across seven data sources: UCI Adult (45,222 rows); ACS Income [7] across thirty-two US states and two protected attributes; HMDA mortgage records for two states, by race and sex and split by loan purpose; COMPAS [9] (5,278 rows on the race arm after ProPublica's documented filter); LSAC bar passage (20,798 rows); the Dutch national census of 2001 (60,420 rows); and Taiwanese credit-card default records of 2005 (30,000 rows). A *population* here is one such unit — a state, cohort or census — and *independent* means no person appears in two of them: arms within a population share people, so however many arms a population contributes, it is counted once (Table I).

COMPAS, LSAC and the Dutch census were added to leave the sources the finding was formed in. The Dutch census carries a between-group gap of 0.298, roughly twice Adult's, making it the test of the standard alternative explanation. LSAC was chosen because bar passage is naturally generous (base rate 0.890).

Convention. $y = 1$ is always the favourable outcome. This required inverting COMPAS's recorded target, which encodes recidivism, and declaring *Female* the advantaged group on its sex arm, since women in that cohort reoffend less. Both are enforced by automated checks, because either error produces every directional result sign-inverted without raising.

TABLE I
WHICH POPULATIONS ENTER WHICH ANALYSIS. AN *arm* IS ONE POPULATION UNDER ONE CONFIGURATION; POPULATIONS ARE COUNTED ONCE HOWEVER MANY ARMS THEY CONTRIBUTE.

Analysis	Populations (arms)
Ablation; who pays	Adult (6 methods $\times$ 5 seeds)
Rate–people divergence	10 pops. (19 arms, 2 attributes)
Label route (income cutoffs)	AL sex; AL, MS, OR, UT race (4 each)
Natural transition	HMDA MS+LA (5 loan purposes)
Threshold route, dense sweeps	Dutch, SC, COMPAS, AL, KY (12 pts.)
Tolerance, 25$\times$ range	AL, CT, KY, OR, SC
Boosted trees	the same five states
Intersectional	Adult + 10 populations
Post-processing regime	17 populations ($r = -0.024$)
Attribute-aware direction	27 of 27, of which 9 pre-registered
Relaxed ζ ordering	26 arms from 15 populations
Crossover intervals	4 non-lending + 3 lending estimates
Sealed test 1 (failed)	8 scored + Taiwan, all fresh
Re-seal (holds)	10 fresh ACS states
Selection-rate floor	10 pops. (19 arms)

TABLE II
SIX IN-PROCESSING METHODS ON UCI ADULT, FIVE SEEDS. EVERY ONE REACHES PARITY, AND EVERY ONE SHRINKS THE POOL.

Method	Acc.	DP	EO	DI	Δpool
Unconstrained	0.847	0.186	0.095	0.300	—
ExpGrad (DP)	0.828	**0.018**	0.280	0.895	-20.5%
GridSearch (DP)	0.827	**0.015**	0.304	0.986	-22.1%
Adversarial deb.	0.826	0.020	0.250	0.889	-14.8%
Prejudice remover	0.837	0.065	0.188	0.686	-10.2%
ExpGrad (EO)	0.836	0.107	**0.032**	0.521	-7.9%

Mitigation. `ExponentiatedGradient` [1] at $\varepsilon = 0.01$ unless stated. No model reads the protected attribute at prediction time except where post-processing is explicitly under test.

Exclusion rules. An arm is discarded if the baseline demographic-parity gap is below 0.05, or if the baseline classifier's accuracy falls below $\max(p, 1-p)$ — the accuracy of always predicting the more common label. The second rule matters more than it sounds (Section X).

Accounting. Table I states which populations enter which analysis, because three counts in this project's own history turned out to be arm counts masquerading as population counts, and a reader should not have to reconstruct the denominators.

IV. PARITY IS REACHED BY WITHDRAWAL — IN SOME POPULATIONS

Table II gives the ablation. Every method reaches its target: demographic parity falls from 0.186 to as little as 0.015 for under two accuracy points, and this holds on a linear model and on an adversarial one alike. The mitigation itself works, which is why we treat it as background rather than as a contribution.

What changes alongside is the part the metric does not report: **every one of these methods shrinks the pool**, by 7.9% to 22.1%, and not one of them closes the gap primarily by extending decisions to the disadvantaged group. Eighteen of nineteen survey arms show the same pattern, which suggests this is a property of the populations rather than of any one method.

TABLE III
WHO PAYS, UCI ADULT. THE RATE-LEVEL VIEW AND THE PEOPLE-LEVEL VIEW OF THE SAME INTERVENTION DISAGREE SYSTEMATICALLY.

Method	Share (rates)	Share (people)	Exchange
ExpGrad (DP)	0.575	**0.742**	2.68
GridSearch (DP)	0.575	**0.743**	2.69
Adversarial deb.	0.508	**0.700**	1.96
Prejudice remover	0.498	**0.673**	2.04
ExpGrad (EO)	0.531	**0.660**	1.92

A. Rates understate what is happening to people

The decomposition of [3] splits the closed gap into the part paid by the advantaged group losing ground and the part gained by the disadvantaged group. Measured in *rates*, that split is close to even — 0.50 to 0.58 of the closure comes from withdrawal (Table III).

Measured in *people* the split is not even. Between 0.66 and 0.74 of the individuals whose decision changed are members of the advantaged group losing a favourable one. The reason the two views disagree is that equal-looking rate changes fall on groups of unequal size, so the rate-level view — which is what a fairness dashboard reports — understates the burden counted in individuals, and it does so in every method tested. The divergence between the two views is itself predictable from a cross-flow diagnostic, at $r = +0.885$ across populations.

The **exchange rate** shows the same thing without needing a share: ExpGrad under demographic parity destroys 2.68 favourable decisions for every one it creates.

On HMDA mortgage data, however, the direction reverses. The identical constraint *grows* the pool by 4.3%, at an exchange rate of 0.50 — two favourable decisions created for every one destroyed. What makes this worth noticing is that the parity metric cannot see the difference: it reports 0.018 on the population that destroyed a fifth of its favourable decisions, and 0.010 on the one that created more than it destroyed.

V. WHAT PREDICTS THE DIRECTION

A. A single-factor design

ACS Income's label is "earns more than \$50,000", and that threshold is a benchmark convention rather than a fact about the world, which makes it a knob we can turn. Moving it on one fixed state varies the selection rate while the population, instrument, feature set, group ratio and proxy structure all stay fixed. The change in the pool rises with the baseline selection rate at $r = +0.801$, and at $+0.980$ once the confounded between-group gap is partialled out. The sign also flips along the way: 22 decisions are destroyed per one created at a rate of 0.03, compared with 0.75 at 0.89.

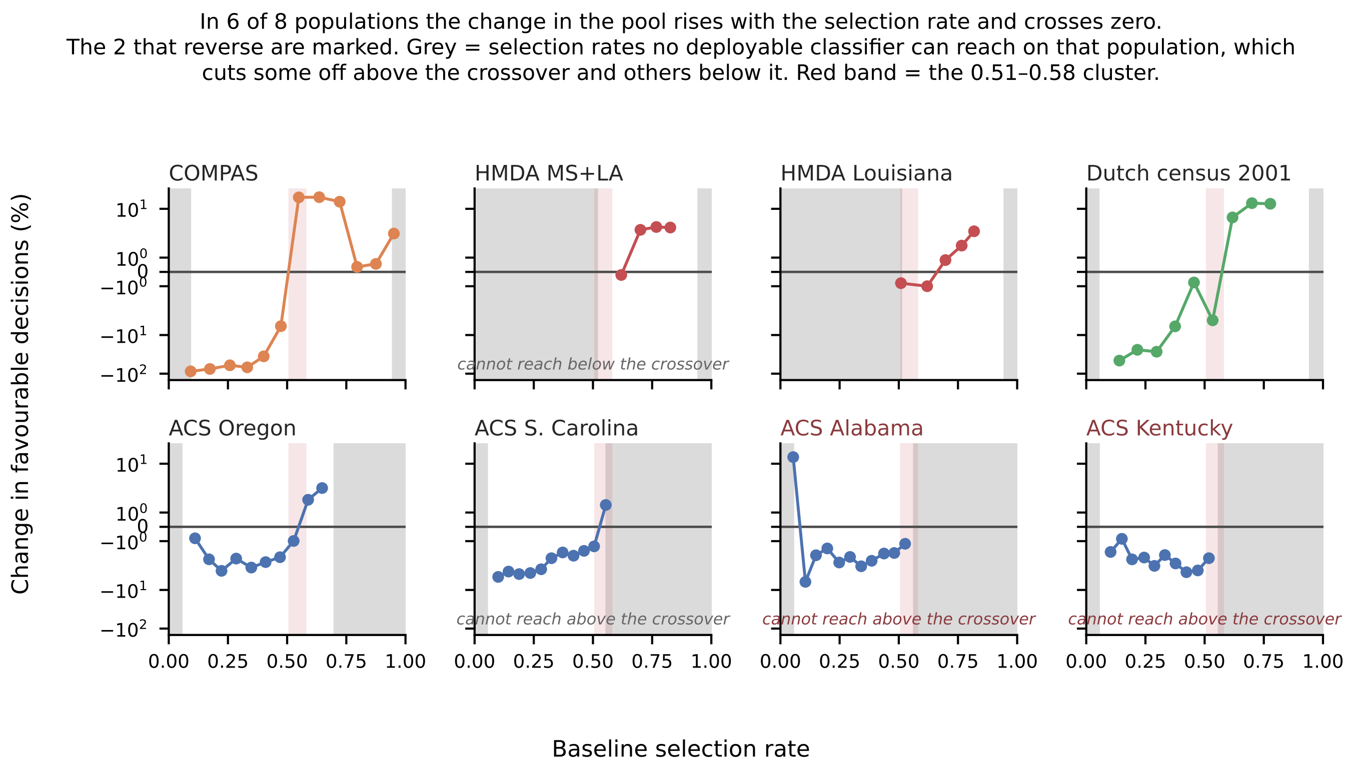

Fig. 1. Every retained arm from every densely swept population. Within a population the change in the pool rises with the baseline selection rate and crosses zero; the two that reverse are marked, and both are sweeps whose viable band confines them below the crossover. Panels are deliberately *not* pooled: across populations a single slope reaches only $r = +0.487$ and fails its pre-registered bar, so one scatter would assert a common curve the data rejects. Vertical band: the 0.511–0.576 cluster of point estimates from Table VII; carrying that table's bootstrap intervals widens it to roughly 0.43–0.58.

B. The transition appears without being manufactured

The same transition also appears where nothing is manipulated at all. Pooling two states across five real loan products, home improvement at a selection rate of 0.555 levels *down* (-1.57%) while refinancing at 0.871 levels *up* ($+2.95\%$), with $r = +0.803$ and Spearman $\rho = +0.900$. This means a bank running one constraint across its loan book would take opportunities away in one product while handing them out in another, and its fairness report would show success in both.

C. It is the rate, not the difficulty of the task

The design above moves the *label*. "Earns over \$100,000" is not a rarer version of "earns over \$20,000"; it is a harder problem. That design therefore cannot separate two explanations which predict everything observed so far.

We separate them by holding the task completely fixed — same rows, features, groups and *fitted scores* — and moving only where the model draws its line. The relationship reproduces, and so does the crossover's location: on Oregon the two routes give 0.362–0.653 and 0.353–0.637. **Task difficulty is not doing the work.**

With twelve points chosen from each population's viable band, the relationship holds on the Dutch census (+0.946), South Carolina (+0.905) and COMPAS (+0.844). Alabama and Kentucky return *negative* values on eleven and ten arms; their viable bands top out at 0.566 and 0.561, at or below every crossover measured, so both sweeps lie almost entirely on one side of the transition. **We therefore claim the relationship across the crossover and withdraw any claim that it holds within the low-rate region alone.**

TABLE IV
THE RELATIONSHIP ACROSS INSTRUMENTS. EACH r IS COMPUTED WITHIN ONE POPULATION OF THAT INSTRUMENT, OVER THE RETAINED ARMS COUNTED IN THE LAST COLUMN — NOT POOLED ACROSS THE INSTRUMENT, FOR THE REASON FIG. 1 DOES NOT POOL.

Instrument	**Domain**	**Population**	r	**Arms**
ACS Income	income	Alabama	+0.801	4
HMDA	mortgages	MS+LA pooled	+0.858	4
COMPAS	crim. justice	race arm	+0.870	5
Dutch census	occupation	sex arm	+0.915	6
LSAC	legal educ.	—	unsweepable	—

D. Five domains, and the group-gap alternative

Fig. 1 shows every arm, and Table IV the summary. The pre-registered alternative — "this is a property of income prediction and American mortgage lending", requiring $|r| < 0.30$ — loses on both new instruments.

The oldest competing account is that the *gap between groups* drives the direction. The Dutch census has a gap of

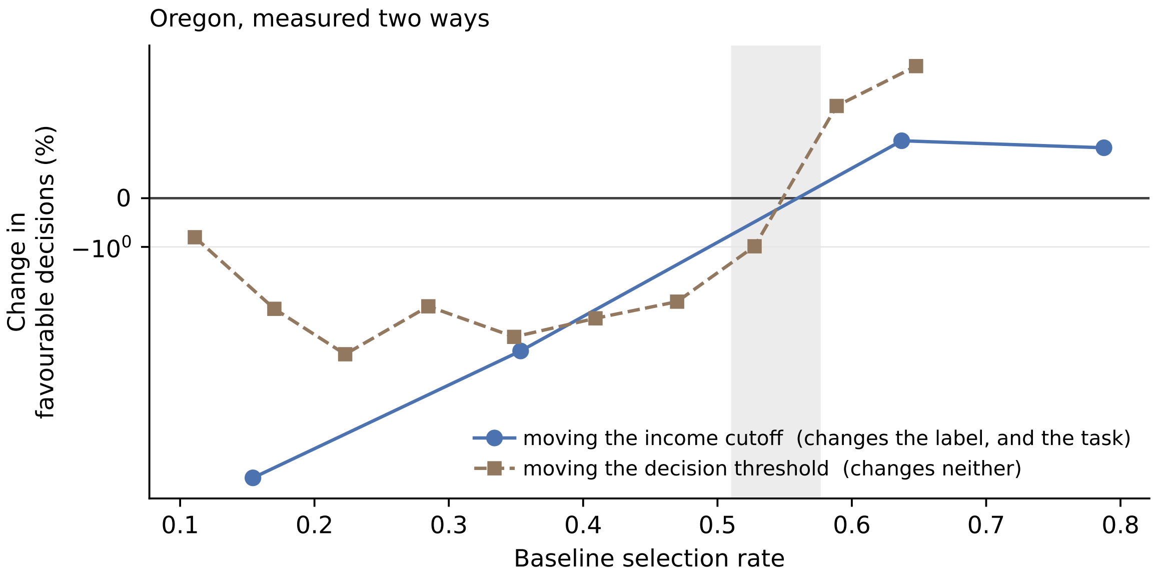

Fig. 2. The design that separates the selection rate from the difficulty of the task. Moving the income cutoff changes the label, and therefore how hard the problem is; moving the decision threshold changes neither, only where the fitted model draws its line. Both routes cross zero inside the same band. If difficulty were doing the work, the second curve would not reproduce the first.

TABLE V
WHAT EACH MANIPULATION PRESERVES

Manipulation	**Direction**	**Magnitude**
Route to the rate	unchanged	differs 3×
Tolerance, 25× range	unchanged	differs 4×
Boosted trees (5/5 pops.)	unchanged	differs 3×
Attribute, sex → race	unchanged	—
Criterion → equalized odds	weaker, unevenly	differs 8×
Optimiser → post-proc.	**vanishes**	—

0.298, roughly twice Adult's, making it the population where that explanation should win. It gives $r = +0.915$ across all six arms, none excluded.

E. Robustness

Table V summarises. **Within a population, and in the attribute-blind regime, the selection rate predicts which way and orders how much. Neither transfers between populations**: the signed distance from a population's own crossover gives ρ of $+0.96$, $+0.95$ and $+0.78$ in three of four populations, but a pooled slope reaches only $r = +0.487$ against a pre-registered $+0.70$ bar.

VI. A Constraint on One Attribute Leaves Subgroups Behind

A constraint is imposed on the attribute someone chose to name. Kearns *et al.* [6] showed that marginal constraints can be satisfied while structured subgroups are badly treated. Table VI is that failure mode measured on Adult.

Constraining sex drives its gap from 0.190 to 0.020 — a success by the certifying metric. The worst *Sex*×*Race* subgroup gap falls only from 0.315 to 0.178, which is **nine times** the gap that was constrained. The subgroup carrying the residual also *changes*, from Female × Black to Male × Black, so an auditor tracking the originally worst-off cell would see improvement while a different cell absorbed the cost.

TABLE VI
UCI ADULT. CONSTRAINING SEX TO 0.020 LEAVES THE WORST *Sex*×*Race* SUBGROUP GAP AT 0.178 — NINE TIMES LARGER — AND MOVES WHICH SUBGROUP IS WORST OFF.

Arm	**Sex**	**Subgroup**	**Worst off**
Unconstrained	0.190	0.315	F×Black
Constrained on sex	**0.020**	**0.178**	M×Black
Constrained on both	0.016	0.048	F×White

This replicates across populations and is more severe than Adult suggests: 13.2× in Mississippi. It is bounded, however, by how large the minority is — the effect scales with minority share at $r = -0.671$, and does not appear at all below a threshold. We present this as an empirical replication of [6] and [4] with a boundary attached, not as a new observation.

A measurement hazard worth naming. Roughly five subgroups per population are too small to support any estimate. A rate with an empty denominator must return undefined rather than zero, or an unmeasurable subgroup masquerades as a perfectly fair one — which is the failure this analysis is most likely to produce and hardest to notice.

VII. The Regime Boundary

The last row of Table V looks like a failure of robustness, but we read it differently, because it falls exactly on a line the theory draws. Post-processing reads the protected attribute at prediction time while the reduction does not, and that is the distinction on which [5] is built.

Running post-processing at each population's natural operating point across 17 populations gives $r = -0.024$, compared with $+0.585$ for the reduction on the same populations. Our pre-registered prediction — that the rule would carry over — failed, and its naive alternative won outright, which is what pointed us at the regime distinction in the first place.

In the attribute-aware regime the theory says there is nothing to predict, because the direction is determined. Its prediction for that regime is that the advantaged group loses and the disadvantaged gains, always. On our populations that holds in **18 of 18**, without exception. That check was post-hoc, so we repeated it as a confirmatory test on nine populations never previously post-processed, with the bar set at all nine and committed before the runs: it holds **9 of 9**. The regime result therefore stands at 27 of 27, nine of them predicted in advance.

A word on what we do and do not claim against the theory. Its Theorem 3 states conditions over the extrema of $\zeta(x) = (\eta(x) - c)/\nu(x)$, and those extrema are unattainable on real data: ζ diverges as $\nu \to 0$, every dataset has points arbitrarily close to $\nu = 0$, and the two regions' empirical ranges therefore always overlap — on Adult by $[-139, 4.1]$ against $[-39, 6662]$. The conditions are *sufficient, not necessary*, so this does not make the theorem false. It makes it *silent* on every population we measure, which is a claim about applicability and not a refutation.

Two further caveats belong here. Our ν is *estimated*, by a probe predicting the protected attribute from the features; their paper prescribes no estimator, so the evaluation is ours, and a different estimator might behave differently. At the same time, a *relaxed* version of their ordering, replacing the unattainable extrema with 5th/95th percentiles, agrees with the measured direction on **24 of 26** arms, compared with 25 of 26 for the selection rate. Taken together, this suggests the theory captures the relevant regime distinction, and that its stated sufficient conditions are not operationally checkable on the populations studied. The regime distinction itself matters most: within the attribute-blind regime the stated sufficient conditions do not yield an operational predictor of the direction, while the selection rate does, and that difference is something an empirical study can settle where a population-level argument cannot.

VIII. Boundaries

A. The crossover is nearly stable

Table VII separates the two groups deliberately, because averaging them would hide the only disagreement in the data. Four non-lending populations, across three domains and two countries, have overlapping intervals. **Carrying those intervals the cluster spans roughly 0.43–0.58**, not the 0.511–0.576 that point estimates alone suggest: COMPAS's interval is six times wider than the Dutch census's, and Oregon's bracket fails to exist in 23% of seed resamples.

TABLE VII

Crossover location with bootstrap intervals over seeds (2,000 resamples). "Located" is the fraction of resamples in which a sign change could be bracketed at all.

Population	Domain	Crossover (95% CI)	Loc.
Non-lending			
COMPAS	crim. justice	0.511 (0.434–0.524)	83%
S. Carolina	income	0.530 (0.526–0.533)	99%
Oregon	income	0.558 (0.556–0.560)	**77%**
Dutch census	occupation	0.576 (0.572–0.580)	100%
Lending — one market, three overlapping estimates			
HMDA pooled	mortgages	0.660	—
HMDA Louisiana	mortgages	0.659	—
HMDA refinance	mortgages	0.758	—

TABLE VIII

Sensitivity to the parity-gap exclusion. Correlation per population as the threshold is loosened.

Population	0.01	0.02	0.05	0.10
Dutch census	+0.851	+0.908	+0.946	+0.938
COMPAS	+0.844	+0.844	+0.844	+0.871
S. Carolina	+0.012	+0.012	+0.905	+0.849
Oregon	+0.095	+0.095	+0.664	—
Alabama	−0.368	−0.368	−0.368	+0.296
Kentucky	−0.590	−0.590	−0.654	—

Every lending estimate sits above every non-lending one. They come from a *single* market measured three overlapping ways, so "lending is different" is a hypothesis here and not a finding.

We therefore treat a crossover near 0.54 as an empirical prior for auditing, not a theoretically justified constant. Nothing in this work explains why 0.54 rather than some other value, and Section X records an attempt to derive it that failed.

Two population properties appeared to explain the residual, at $r = +0.724$ and $+0.865$. They are collinear with each other at $+0.947$ and neither is distinguishable from chance at four populations ($p = 0.277$, $p = 0.135$). We report them as a hypothesis for a larger study, not a formula.

B. The procedure is unusable on very generous tasks

Varying the selection rate by moving a threshold works by *degrading* the classifier. On a task where most people already qualify, reaching a low selection rate requires a model that refuses qualified people, and past a point that model is worse than a constant. Defining the *viable band* as the range of selection rates reachable by a classifier still beating $\max(p, 1-p)$: Dutch spans 0.895, COMPAS 0.859, typical ACS ≈ 0.51, and LSAC only **0.056**. On LSAC, where 89% pass the bar, no choice of thresholds produces a usable sweep. This is computable before running anything.

When the viable band forbids a sweep, the crossover can still be located by comparing natural sub-populations — which is how the mortgage crossover was first found.

TABLE IX
THE SEALED PREDICTION. POPULATIONS NEVER PREVIOUSLY MEASURED; PREDICTIONS COMMITTED BEFORE THE RUNS.

Population	Rate	Δpool	Pred.	Actual
MO, $100k	0.026	−16.48%	up	**down**
AZ, $100k	0.037	−23.34%	up	**down**
IN, $70k	0.081	−23.32%	up	**down**
TN, $50k	0.255	−1.68%	down	down
MD, $50k	0.508	+0.78%	down	**up**
MI, $30k	0.612	+0.85%	up	up
NC, $20k	0.789	+0.16%	up	up
GA, $10k	0.905	+0.30%	up	up
Taiwan 2005	0.885	+0.68%	up	up

TABLE X
THE RE-SEALED PREDICTION: THE UNREFINED RULE — DOWN BELOW 0.54, UP AT OR ABOVE — COMMITTED BEFORE TEN NEVER-MEASURED STATES RAN. BAR: 9 OF 10 *and* BEAT THE BEST CONSTANT.

Population	Rate	Δpool	Pred.	Actual
NE, $100k	0.014	−6.35%	down	down
AR, $50k	0.195	−2.55%	down	down
OK, $50k	0.220	−12.78%	down	down
WA, $70k	0.233	−5.52%	down	down
NV, $50k	0.297	−1.57%	down	down
MN, $30k	0.699	−0.04%	up	**down**
CO, $30k	0.701	+0.63%	up	up
KS, $20k	0.767	+0.58%	up	up
WI, $20k	0.814	+0.69%	up	up
IA, $10k	0.896	+0.04%	up	up

C. A sealed prediction, and what it cost

Every result above is a correlation computed after the arms were run, which is the weakest support for a claim of the form *you can know in advance*. We therefore ran the strongest format available: nine populations whose direction had never been measured, with the rule, the populations and the pass mark committed before any of them existed. Eight enter the sealed score; the ninth, Taiwan, was sealed alongside them but registered as reported separately, because a single non-Western population can carry no claim on its own (it was predicted correctly: rate 0.885, observed $+0.68\%$).

It failed. Four of eight, against a bar of seven, not beating a constant.

What failed is a refinement, not the rule. Table VIII shows the exclusion threshold doing real work: South Carolina moves from $+0.905$ to $+0.012$ when it is loosened. Investigating that, we found the newly admitted arms are not noise — at twenty seeds they are consistently *positive*, $+8.73$, $+6.85$, $+9.09$, six standard errors from zero — and concluded the relationship turns back up below a selection rate of 0.10. The sealed rule carried that refinement.

All three held-out low-rate arms came out strongly *negative*. Scoring the same eight arms under the simpler rule we held before the refinement — down below the crossover, up above it — gives **seven of eight**, its only miss being Maryland at 0.508, within 0.03 of the crossover on an effect of $+0.78\%$. That seven is post-hoc and we do not claim it; what was sealed scored four.

The refinement was route-specific. Its arms reach a rate near 0.05 by thresholding one fixed score vector; the sealed arms reach it with a genuinely rarer label. At that rate the two disagree completely, $+8.73$, $+6.85$, $+9.09$ against -16.48, -23.34, -23.32. So Section V-C qualifies: the two routes agree through the middle of the range and diverge at the bottom, and the accuracy rule does not catch the divergent arms — they clear the trivial-predictor bar and remain artifacts.

A refinement drawn from four in-sample populations, internally consistent and backed by a twenty-seed replication, made the prediction worse than the rule it replaced and no better than a constant. Nothing in the in-sample data could have shown that.

D. The rule re-sealed, and it holds

After that failure, the only move that could settle anything was to seal the *unrefined* rule — down below 0.54, up at or above, with no low-rate clause — against populations with no measurement history of any kind. We used ten ACS states this project had never touched, one arm per state so that the arm count is the population count by construction, with income cutoffs assigned to spread the expected rates across both sides of the crossover so that no constant could mimic the rule. The 0.54 prior itself predates the test: it is the mid-point of the non-lending cluster, fixed from four populations that include none of the ten. The bar was committed with the populations before any arm ran: at least 9 of 10, *and* strictly beating the best constant.

It holds: 9 of 10, best constant 6 (Table X). The prediction used the baseline selection rate and the fixed 0.54 prior alone — no sweep, no per-population characterisation. Below 0.10, where the refuted refinement said up, Nebraska at 0.014 moved down by 6.35%, as the unrefined rule predicts.

The miss is scored by the criterion sealed with it, and it is *not* excused as a boundary call: Minnesota sits at 0.699, far from the crossover, so under the registered rubric it counts against the rule itself. What can be said post-hoc is that its effect is the smallest in the set by an order of magnitude and flips sign across seeds (mean -0.04%) — a rule predicting a sign has nothing to grip on an arm whose true value may not have one, while Colorado, 0.002 away in rate, moved $+0.63\%$ and was called correctly. The registered rubric did not anticipate near-zero magnitudes away from the boundary; a future seal should state a minimum-magnitude guard in advance, and this one is charged the miss for not having done so.

E. Small populations

Below roughly 2,500 test subjects the method's own randomness exceeds the entire effect of the constraint. This is a specific instance of predictive multiplicity [12] — models with equivalent aggregate performance disagreeing substantially on individuals — and we position it as a caution supported by that literature rather than as a novel finding. COMPAS has 1,584 and flips sign between seeds on two of four arms; LSAC has

6,240 and flips on none. COMPAS carries direction here and never magnitude.

IX. What to Do About It

Algorithm 1 states the procedure. Three of its steps are worth drawing out, because each is a place where a plausible-looking run returns a meaningless answer.

It can refuse. Lines 6–8 compute the viable band and exit before any constraint is fitted, if no threshold reaches a low selection rate with a model still beating the trivial predictor. On LSAC that band spans 0.056 and the procedure declines, rather than returning arms that would be discarded later.

It can return VOID. Line 16 requires the pool to have moved at all. A correlation computed over arms that barely move is fitted to noise: Connecticut returns $r = -0.924$ on a spread of a tenth of a percentage point, and without this test that number reads as a refutation of the whole finding.

It returns a sign, never a size. The distance from the crossover *orders* the effect within a population, so a team can rank its own segments; but no slope transfers between populations, so lines 25 and 27 return a direction and stop there.

The floor is **not a new method**: a selection-rate floor is a linear moment constraint inside the framework of [1], and a variant of the minimum rate constraints of [2]. It costs about 0.12 accuracy points and moves the exchange rate from 1.47 favourable decisions destroyed per one created to 0.88 across nineteen arms, landing below 1.0 in sixteen of them, with the benefit scaling with the damage at $r \approx -0.99$.

X. Method, and What We Got Wrong

Every prediction was registered with its thresholds *and the naive alternative it had to beat*, before the data existed. Table XI is the complete ledger — every registered or sealed test this work ran, with its committed bar and its outcome, failures included and uncorrected. The rows should not be read as seven attempts at one claim, of which two succeeded: each test targeted a *different* claim, each failure removed a stronger claim than the one that survives, and the last column records what each row settled. The failures are boundary measurements; the passes are tests of the claim *as those boundaries define it*. Three episodes matter for reading the results above.

A held-out prediction, scored as a failure. The clearest evidence format available is a prediction registered before the data exists. We ran one: after adding the accuracy rule below, four HMDA populations the sweep had never touched were swept and scored under it, with the prediction that each would show the relationship, clear the spread guard and bracket a crossover. **Two of four did**, so the test is scored a failure. Its named alternative — that lending is a domain where this route does not work — nonetheless loses on all four, with $r \geq +0.80$ throughout and $+0.995$ on refinance. Three independent estimates of the lending crossover agree (Table VII).

A pre-registered test that failed. The rule does not survive equalized odds at the bar we set: $r = +0.644$ on two states,

Algorithm 1 Directional audit: decide whether the predictor applies at all; if it does, locate the crossover and decide whether to add a floor

Require: deployed model h, held-out (X, y, a), fairness constraint C
Ensure: predicted direction, and whether a selection-rate floor is needed
1: **if** C does not constrain selection rates **then**
2: **return** OUT OF SCOPE {e.g. equalized odds}
3: **end if**
4: $\pi \leftarrow \max(\bar{y}, 1 - \bar{y})$ {trivial-predictor accuracy}
5: $s \leftarrow h(X)$
6: $\mathcal{V} \leftarrow \{\tau : \mathrm{acc}(s > \tau) \geq \pi \ \wedge\ 0.05 \leq \overline{(s > \tau)} \leq 0.95\}$
7: **if** $\mathrm{span}(\mathcal{V}) < 0.40$ **then**
8: **return** UNSWEEPABLE; compare natural sub-populations
9: **end if**
10: $T \leftarrow 12$ thresholds from $\mathcal{V}$, spread evenly in *selection rate*
11: **for all** $\tau \in T$ **do**
12: $r_\tau \leftarrow \overline{(s > \tau)}$; fit C on the arm thresholded at τ
13: $\Delta_\tau \leftarrow$ % change in favourable decisions
14: discard τ if parity gap < 0.05 or $\mathrm{acc} < \pi$
15: **end for**
16: **if** $\max \Delta - \min \Delta < 2$ points **then**
17: **return** VOID {nothing moved; not a refutation}
18: **end if**
19: $c \leftarrow \big(\max\{r_\tau : \Delta_\tau < 0\},\ \min\{r_\tau : \Delta_\tau > 0\}\big)$
20: **if** c is empty **or** c is far from 0.54 **then**
21: inspect the sweep before believing it
22: **end if**
23: $r_0 \leftarrow$ selection rate of the deployed model
24: **if** $r_0 < \min c$ **then**
25: **return** WITHDRAWAL; add a selection-rate floor
26: **else**
27: **return** EXTENSION; the floor buys little
28: **end if**

$+0.334$ on five, against a $+0.70$ threshold. Alabama alone would have given $+0.822$ with every prediction holding.

Two headline results withdrawn. The operating-point design varies the selection rate by degrading the classifier, and nothing bounded how far. Excluding arms beaten by the trivial predictor leaves Alabama with two arms and LSAC with one: $r = +0.979$ and $r = +0.968$, both previously reported, are void. The same exclusion *rescued* mortgage lending, which had failed outright.

A derivation that a constant beat. The most obvious extension of this work is a theoretical account of *why* the selection rate should matter. A general theory already exists and we cite it: [5]'s Theorem 3 gives explicit conditions, over the extrema of the score-region quantity $\zeta(x) = (\eta(x) - c)/\nu(x)$, under which every group's rate weakly falls or weakly rises. Those conditions are ordering relations rather than anything a practitioner can compute, and Section VII reports that they cannot be evaluated on real data at all — while a relaxed form

TABLE XI

EVERY REGISTERED OR SEALED TEST. EACH ROW TARGETS A DIFFERENT CLAIM; THE LAST COLUMN IS WHAT ITS OUTCOME SETTLED. NOTHING IS OMITTED.

Test	**Bar / outcome**	**What it settled**
HMDA sweep	4 of 4; got 2 — fail	our sweep procedure is unreliable on mortgage data, so no claim rests on it there
EO transfer	$r \geq +0.70$; $+0.334$ — fail	the rule works for parity, which controls approval rates directly; it failed for equalized odds, which does not
Post-processing transfer	carry over; $r = -0.024$ — fail	the rule stops where the model may read the protected attribute — the boundary of Sec. VII
Derivation	beat a constant; beaten — fail	we cannot explain *why* the rule works, and claim only that it does
Sealed rule, refined	7 of 8; got 4 — fail	the extra clause added from early data was wrong, and is withdrawn
Attribute-aware replication	9 of 9; got 9 — **holds**	where the attribute is readable, the disadvantaged group always gains — as theory says, predicted in advance
Re-seal, unrefined rule	9 of 10 + beat constant; got 9, constant 6 — **holds**	the simple rule predicts populations it has never seen, from the rate alone

of the same ordering tracks the direction on 24 of 26 arms.

We attempted the missing step — a derivation predicting the sign from the selection rate directly — and pre-registered its bars. **It cleared them and was then beaten by a constant**, 12 correct against 13 out of 14 held-out arms. That failure is why this paper reports an empirical regularity rather than a mechanism, and why we do not attempt a fourth derivation here.

A plausible result that was arithmetic. A post-processing comparison appeared to show a clean reversal. It was void: the method re-derives its own thresholds and returned an identical model at all six operating points, so the measured effect was a constant divided by a shrinking baseline. What exposed it was an exchange rate of 0.000.

The claim that survives every test above, stated once and in full. Within a population, in the attribute-blind in-processing regime, across the transition: the baseline selection rate, read against a crossover prior near 0.54, predicts the direction in which a parity constraint moves the pool of favourable decisions — sealed at 9 of 10 on populations never previously measured — and orders the magnitude within that population. The boundaries, equally part of the claim: the crossover clusters at 0.43–0.58 and is unexplained; magnitude does not transfer across populations and no pooled slope exists; the deepest-tail arms of a threshold sweep are unreliable in direction, so at low rates only the label route and natural sub-populations carry the claim; and in the attribute-aware regime the direction is determined by theorem, the rate predicts nothing, and there is nothing left to predict.

XI. LIMITATIONS

The sealed record is one failure and one pass. The first sealed test (Section VIII-C) failed at four of eight, carrying an in-sample refinement; the re-seal of the unrefined rule (Section VIII-D) passed at nine of ten with the bar at nine, beating the constant at six. The pass has its own limits: all ten populations share one instrument and one country — held-out ACS states are the cheapest clean populations available, so the sealed evidence buys depth, not breadth — and the single miss is charged to the rule by the registered rubric, because no minimum-magnitude guard was stated in advance.

The two manipulations diverge in the score tail. The deepest-tail arms of an operating-point sweep are unreliable in direction at whatever rate they land (observed from 0.027 to 0.142): they close the gap by *lifting* where every natural-route arm at a matched rate closes it by *cutting*, and the accuracy rule does not separate them — all fifteen such arms clear it. What selects lift over cut is measured to be none of the rate, the gap, the qualified reservoir, or the threshold height; it remains open. We claim the relationship across the transition, on the label route and natural sub-populations.

Magnitude does not transfer between populations, only within them. **The crossover clusters but is not explained**: two candidate predictors are collinear at $+0.947$ and neither survives four populations. **Coverage** is seven data sources, three countries and two decades; Taiwan is the only non-Western population, and one population is not coverage. **The regime result** was post-hoc at 18 of 18 and has since been confirmed pre-registered at 9 of 9, but it confirms a published theorem rather than establishing one. **The remedy is a variant**, not a discovery.

XII. DISCUSSION

The finding a practitioner can use is small and specific: **look up the rate at which your model currently says yes, locate your own crossover, and you can anticipate whether a parity constraint is about to give or to take.** We do not claim that the selection rate *causes* the direction; we claim that it predicts it, within a population, in the attribute-blind regime, and with the boundaries of Section VIII attached.

What makes this worth stating is not that the rule is surprising once seen, because it is not. It is that the certifying metric — the number a deployment team reports, an auditor checks, and a regulator may one day require — comes out identical in both directions. A constraint that withdraws a fifth of all favourable decisions and one that creates more than it destroys produce the same certificate, so nothing in the existing reporting pipeline would ever distinguish them.

The literature already knew the metric was silent about this. What was missing, and what this paper adds, is a way to know in advance — from a number a team already has — which of the two outcomes its own system is heading toward.

REPRODUCIBILITY

Every load-bearing figure is re-derived from stored results by an automated check that fails if a document and its data disagree; 55 checks across five suites accompany this work, 28 of them of that document-against-data form. Pre-registrations are committed before the runs they govern.

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